DOCUMENT 04

# System Workflow Specification

End-to-end flows, states, retries, and edge-case behavior

## Study Topic Extractor

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*Turn long study material into a focused study roadmap.*

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# 1. Purpose and Conventions

This document defines how data and control move through Study Topic Extractor for each user-visible flow. Stage names match the pipeline modules in the NLP/AI Specification. Flows marked (P2) are Phase 2.

| Term | Definition |
|---|---|
| Analysis | One execution of the pipeline on one text with one configuration. Produces an Analysis Result. |
| Topic | A ranked study concept (word or phrase) with score, priority, summary, keywords. |
| Keyword | A distinctive term or phrase used for scanning vocabulary; topics may be keywords but not all keywords are topics. |
| Score | Importance value 0 to 100 computed from weighted signals. |
| Priority | HIGH, MEDIUM or LOW derived from Score using configurable thresholds. |
| Study Roadmap | The complete structured output shown on the Results page. |
| Extraction quality | 0 to 1 estimate of how much readable text a PDF yielded. |

# 2. Flow 1: Text Analysis

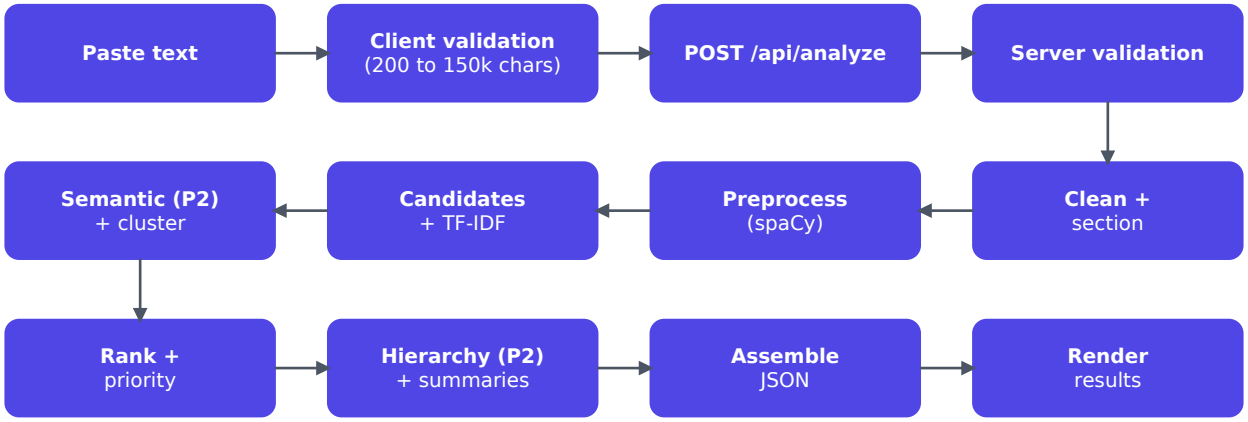



*Figure 1. Text analysis flow*

## 2.1 Step Detail

| # | Step | Actor | Description | Failure handling |
|---|---|---|---|---|
| 1 | Input | User | Pastes or types text in the editor; optional title. | Counter turns red beyond limit; button disabled. |
| 2 | Client validation | Frontend | Trim, check length bounds, strip null characters. | Inline error; no request sent. |
| 3 | Request | Frontend | `POST /api/analyze` with `{text, title?, options?}`. | Network failure: toast + Retry (see 8.1). |
| 4 | Server validation | API | Pydantic validation, length bounds, language sanity check. | TEXT_TOO_SHORT or DOCUMENT_TOO_LARGE (422/413). |
| 5 | Cache lookup | API | Key = sha256(normalized text) + sha256(config). Hit returns instantly. | Miss continues. |
| 6 | Pipeline | Services | Clean, section, preprocess, candidates, TF-IDF, keywords, rank, priority, summaries. Phase 2 adds semantic, cluster, hierarchy, graph. | Exception becomes PROCESSING_FAILED with request_id. |
| 7 | Response | API | Analysis Result JSON (see Result Schema). | - |
| 8 | Display | Frontend | Store in `sessionStorage`, route to `/results/{analysis_id}`. | Corrupt storage: show error state and offer re-run. |

## 2.2 Result Schema (abridged)

```
{
  "analysis_id": "b7e1c0de-...",
  "document": {"id": null, "title": "Machine Learning Notes", "source_type": "text",
               "char_count": 5231, "word_count": 812, "section_count": 4, "language": "en"},
  "stats": {"topics_found": 18, "high": 6, "medium": 8, "low": 4, "processing_ms": 2140},
  "config": {"weights": {"tfidf": 0.47, "frequency": 0.29, "position": 0.24, "semantic": 0.0},
             "thresholds": {"high": 80, "medium": 50}, "semantic_enabled": false,
             "max_topics": 25, "pipeline_version": "1.0.0"},
  "topics": [
    {"id": "t_001", "name": "Machine Learning", "normalized": "machine learning",
     "rank": 1, "score": 95.0, "priority": "HIGH", "ngram_size": 2, "occurrences": 14,
     "first_position_pct": 0.01,
     "components": {"tfidf": 0.93, "frequency": 1.0, "position": 0.88, "semantic": null},
     "summary": "Machine learning is a branch of artificial intelligence ...",
     "definition": "Machine learning is a branch of artificial intelligence ...",
     "keywords": ["training", "model", "data"],
     "parent_id": null, "children_ids": [], "cluster_id": null}
  ],
  "keywords": [{"term": "labeled dataset", "score": 0.71, "count": 5}],
  "hierarchy": null,
  "graph": null,
  "warnings": []
}
```

# 3. Flow 2: PDF Analysis

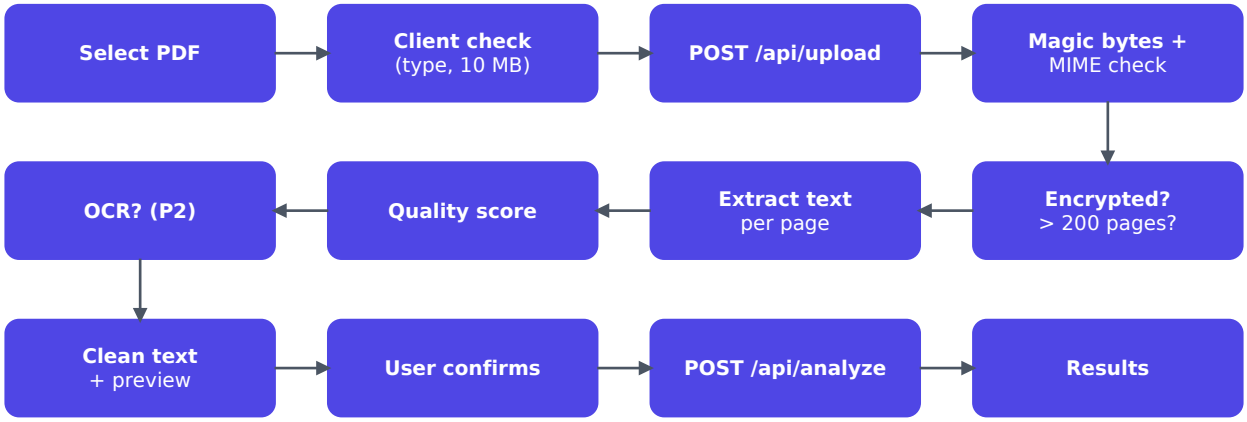



*Figure 2. PDF analysis flow*

## 3.1 Rules

- The upload endpoint returns extracted text, page count and quality; it does not analyze. This lets the user preview and lets Flow 1 run unchanged. The frontend may auto-continue when quality >= 0.5.
- **Extraction quality:** `quality = readable_pages / total_pages` where a page is readable if it has at least 50 characters and at least 60% alphabetic characters after cleaning.
- **Quality < 0.5 but > 0:** return text with warning `LOW_EXTRACTION_QUALITY`; UI shows a dismissible banner suggesting OCR (P2) or pasting text.
- **Quality = 0 or no text:** MVP returns `NO_READABLE_TEXT`. Phase 2 with `ENABLE_OCR=true` attempts OCR at 200 DPI on pages without text, capped at 50 pages, then re-scores.
- **Encrypted PDF:** `PASSWORD_PROTECTED_PDF`. The product never asks for or stores PDF passwords in MVP.
- **Page or text limit exceeded:** `DOCUMENT_TOO_LARGE`. Suggest splitting the document.
- **Temporary data:** PDF bytes are held in memory and released when the request ends. (P2, authenticated save: bytes go to private storage.)

## 3.2 Sequence

```
Browser              Next.js            FastAPI               Services
  | select file         |                   |                     |
  |-------------------->| validate type,size|                     |
  |                     |--POST /api/upload>| verify magic bytes  |
  |                     |                   |--extract(bytes)---->| PyMuPDF
  |                     |                   |<--text,pages,quality|
  |                     |<--200 {text,...}--|                     |
  |<--preview + quality-|                   |                     |
  | click Extract Topics|--POST /api/analyze>|--run_pipeline----->| NLP stages
  |                     |<--200 result------|<--result------------|
  |<--navigate results--|                   |                     |
```

# 4. Flow 3: Exam Focus Mode (Phase 2)

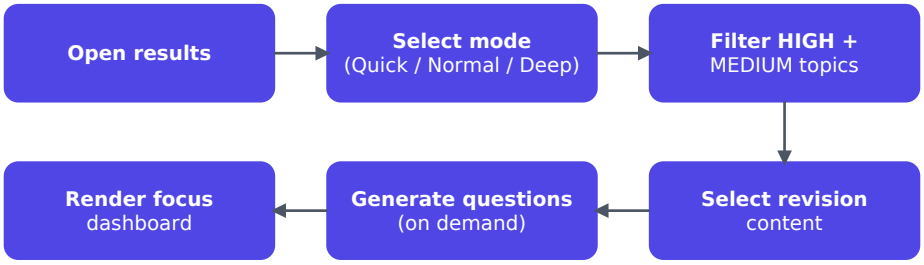



*Figure 3. Exam Focus Mode flow*

| Mode | Data required | Source |
|---|---|---|
| Quick Revision | HIGH topics, keywords, one-line definitions | Already in Analysis Result |
| Normal Study | Topics, subtopics, summaries, keywords, priority | Analysis Result (hierarchy required for subtopics) |
| Deep Study | Above + definitions, relationships, source sentences, possible questions | Analysis Result + POST `/api/documents/{id}/questions` (or `options.include_questions=true`) |

Question generation runs on demand, only for HIGH topics by default (max 6 per topic), and is cached per analysis run. Template-based generation works offline; LLM enhancement is optional and falls back to templates on error or timeout (8 s).

# 5. Flow 4: Export

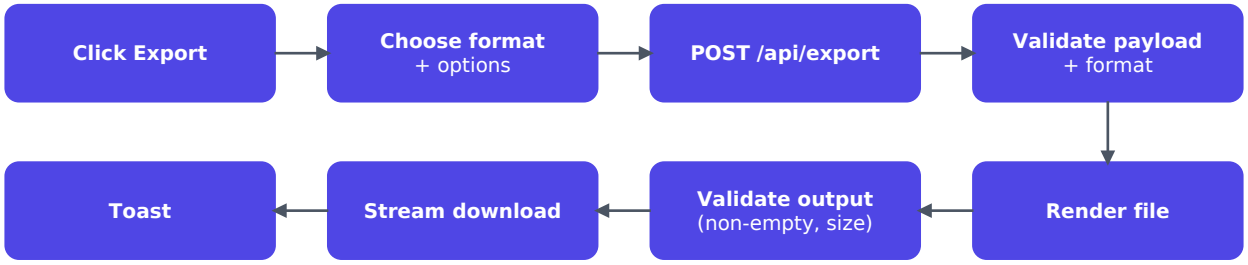



*Figure 4. Export flow*

| Format | Content | Notes |
|---|---|---|
| PDF | Title, overview, topics with score, priority, summary, keywords; keyword list; (P2) hierarchy | ReportLab, A4/Letter, uses fonts embedded; max 60 pages |
| CSV | Columns: rank, topic, score, priority, keywords, summary, occurrences | UTF-8 with BOM for Excel; formula-injection guard (prefix ' on cells starting with = + - @) |
| TXT | Plain-text roadmap grouped by priority | UTF-8, LF line endings |
| MD (optional) | Headings by priority, bullet topics | - |
| JSON (optional) | The Analysis Result | - |

# 6. Loading, Empty, and Error States

| Situation | Behavior | User message |
|---|---|---|
| Loading (analysis) | Stage-based progress; Cancel available; UI never blocks | Finding topics... |
| Slow (> 15 s) | Hint appears; P2 offers background processing | Large documents can take longer. |
| Empty document | Blocked client-side and server-side | Please provide at least 200 characters of study material. |
| No topics found | Result with zero topics and warning `NO_TOPICS_FOUND`; empty state with tips | We couldn't identify clear topics. Try adding more detailed notes. |
| Invalid file | Reject before upload where possible | Please upload a valid PDF file. |
| No readable text | Suggest paste or OCR (P2) | We couldn't find readable text in this document. |
| Too large | Reject; suggest splitting | This document is larger than the supported processing limit. |
| Processing failure | Show Retry; log request_id | We couldn't analyze this document. Please try again. |
| Rate limited | Countdown from `Retry-After` | Too many requests. Please wait a moment and try again. |

## 6.1 Retry Behavior

- **Automatic:** GET requests retry up to 2 times with exponential backoff (500 ms, 1.5 s) on network errors or 502/503/504. POST analyze, upload and export are not retried automatically.
- **Manual:** every error state offers Retry that re-sends the original payload held in memory.
- **429:** the client waits for `Retry-After` seconds before re-enabling the button.
- **Idempotency:** analysis is deterministic and cacheable, so repeating a request is safe.

# 7. Edge-Case Behavior Matrix

| Case | Detection | System behavior |
|---|---|---|
| Extremely large documents | Length and page checks | Reject above limits. P2: 30k to 150k chars run as async job with polling. |
| Very short text | < 200 chars | TEXT_TOO_SHORT. |
| Short but valid (200 to 600 chars) | Length | Run with warning `SHORT_DOCUMENT`; TF-IDF blends background IDF; fewer topics returned. |
| Poor-quality PDF | Quality < 0.5 | Warning; results still produced. |
| Corrupted PDF | fitz open error | INVALID_PDF. |
| Password-protected PDF | `needs_pass` | PASSWORD_PROTECTED_PDF. |
| Scanned PDF | No text layer | MVP NO_READABLE_TEXT; P2 OCR. |
| Duplicate document (P2) | Same content hash for user | Return existing document and latest run; option to re-analyze. |
| Non-English text | Stopword-ratio language check | Warning `LOW_LANGUAGE_CONFIDENCE`; proceed with English pipeline. |
| Math-heavy content | High ratio of symbols | Symbols stripped from candidates; warning `MATH_CONTENT`. |
| Repeated content | Duplicate sentence detection | Duplicates collapsed before counting. |
| No identifiable topics | Zero candidates pass filters | Return empty topics, warning, empty state. |
| User navigates away mid-analysis | AbortController | Request cancelled; server work may finish but result is discarded. |
| Browser refresh on results | sessionStorage present | Results reloaded from session; otherwise prompt to re-run. |

# 8. Data Flow Summary and Persistence (Phase 2)

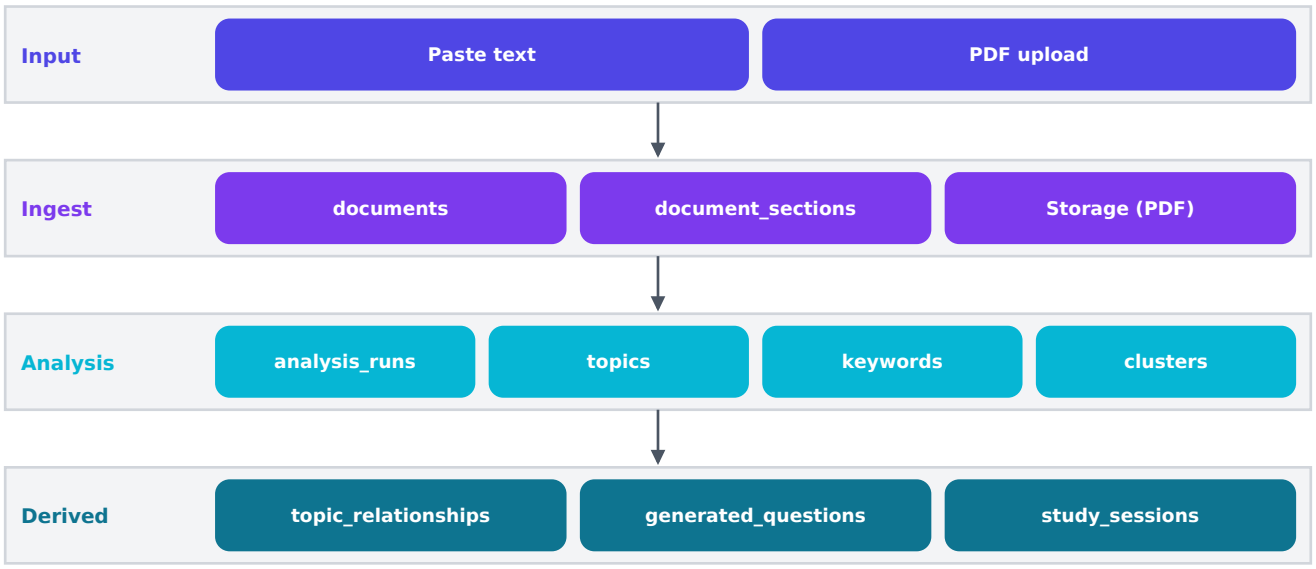



*Figure 5. Persisted data flow (Phase 2)*

Persistence only happens for authenticated users who opt in (`save=true`). Deleting a document cascades to sections, runs, topics, keywords, relationships, questions and sessions, and removes the stored PDF.

# 9. Acceptance Criteria

1. Each flow behaves as specified for success and each listed failure.
2. Errors use the catalog messages exactly.
3. Cancel in Processing State aborts the network request within 200 ms.
4. PDF flow never writes user files to disk in MVP.
5. CSV export neutralizes formula injection.
6. Repeated identical analysis requests return identical topics and scores.